Token Compression for Video-LLMs

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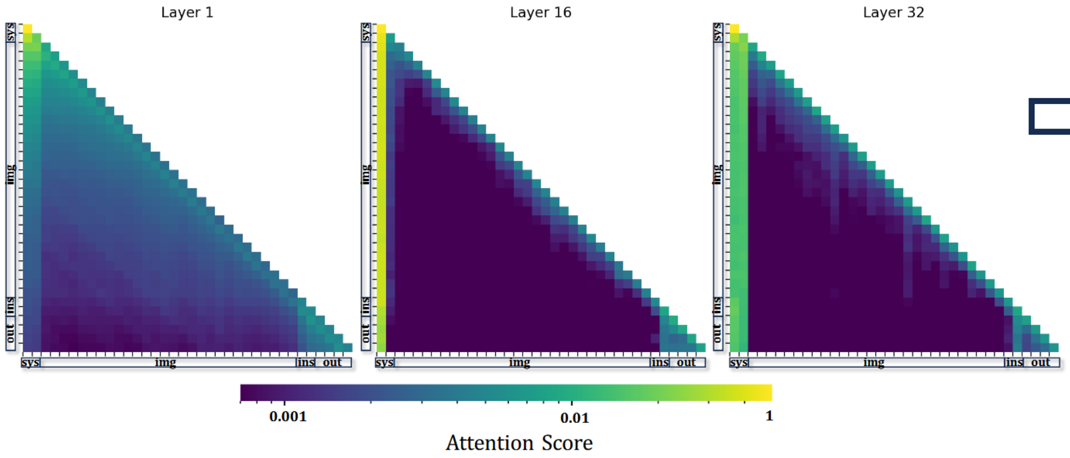
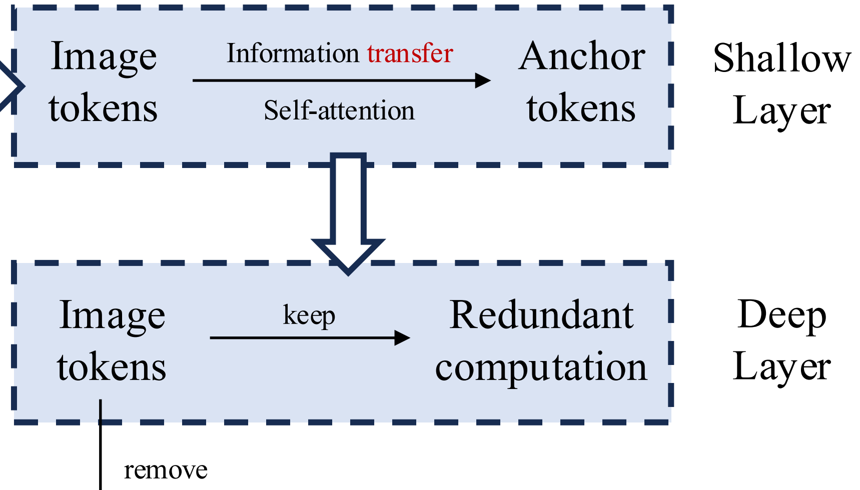
2026.05.06

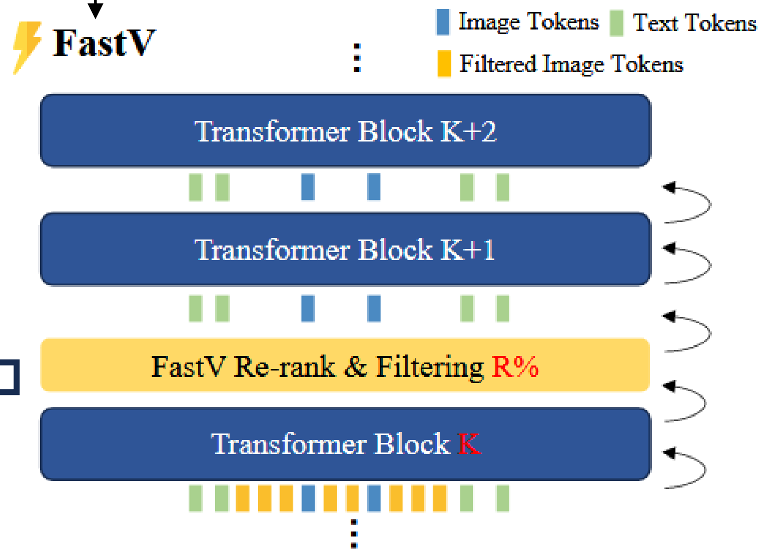
An Image is Worth 1/2 Tokens After Layer 2: Plug-and-Play Acceleration for VLLM Inference



Insight



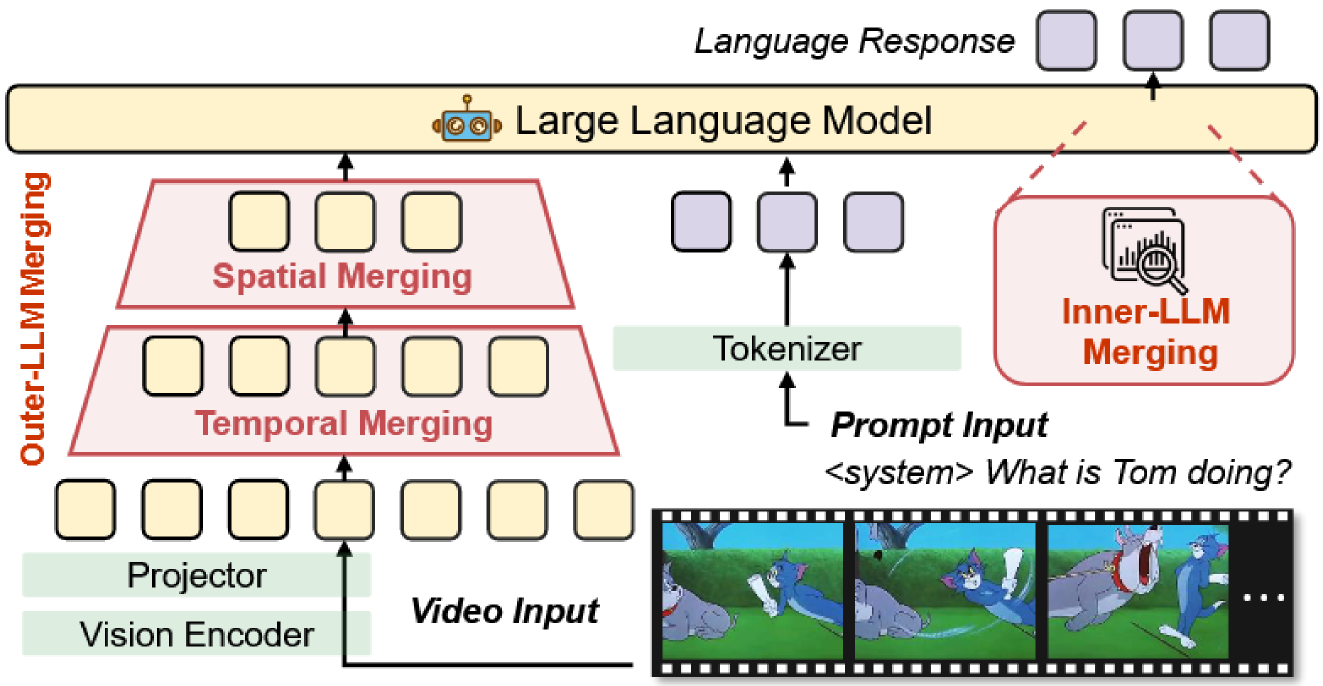
- Shallow Layer: Attention distributes relatively smooth across different type of tokens
- Deep Layer: Attention scores are aggregated to sys, ins and out tokens and attention over image tokens is rather sparse



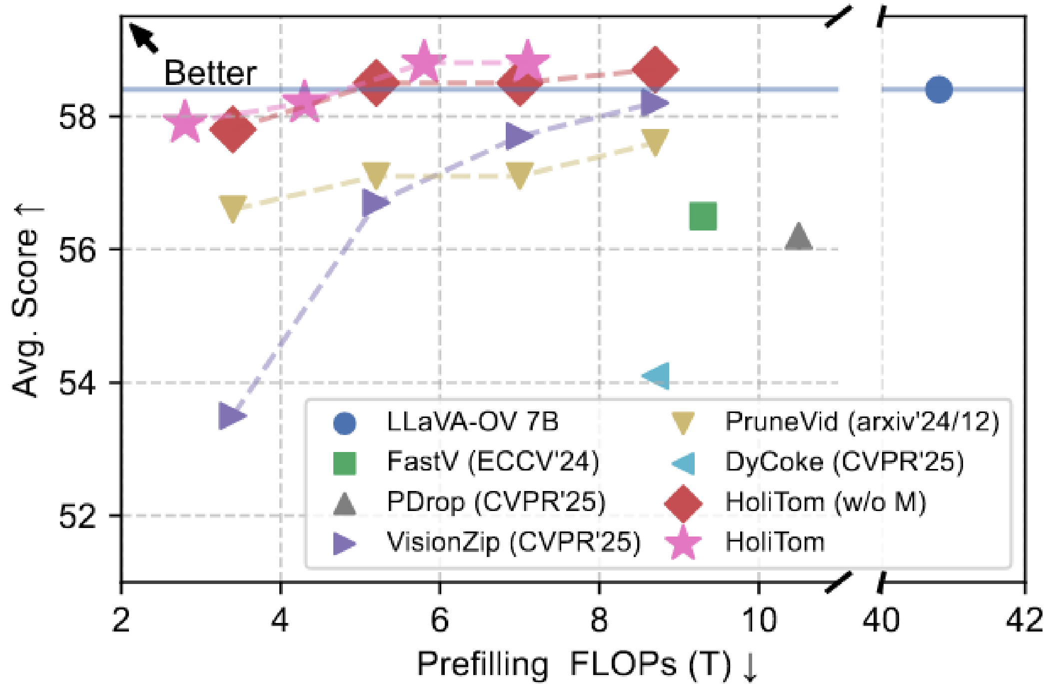
Trade-off

- Inner-LLM methods: incur intrinsic computational overhead in shallow layers
- Discarding image tokens rather than merging them may lead to information loss, potentially resulting in fatal failures when encountering corner or rare cases in real-world applications

HoliTom : Holistic Token Merging for Fast Video Large Language Models

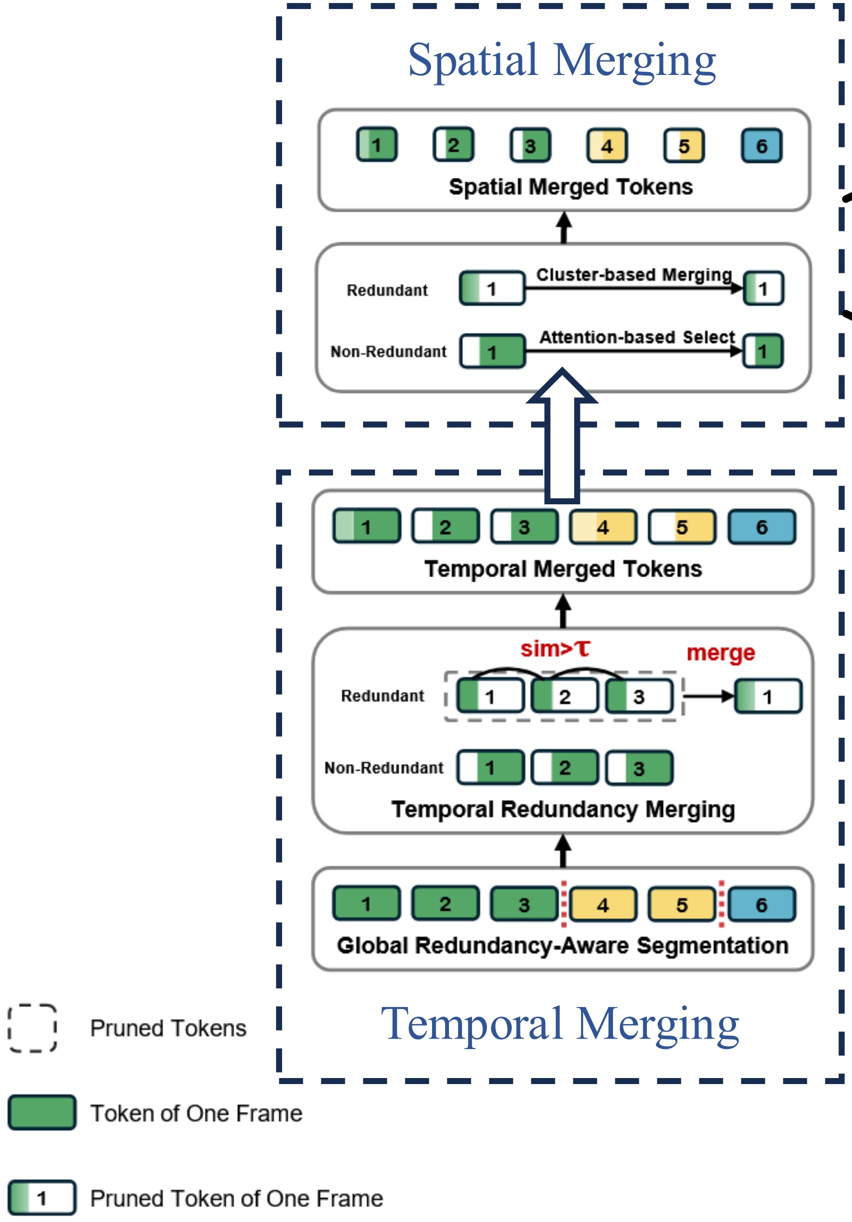


Overview of HoliTom



Trade-off curve

Outer-LLM



Redundant:

- Attention is computed within a single frame rather than across frames
- Merge with DPC-KNN, then concatenate according to their original spatial order

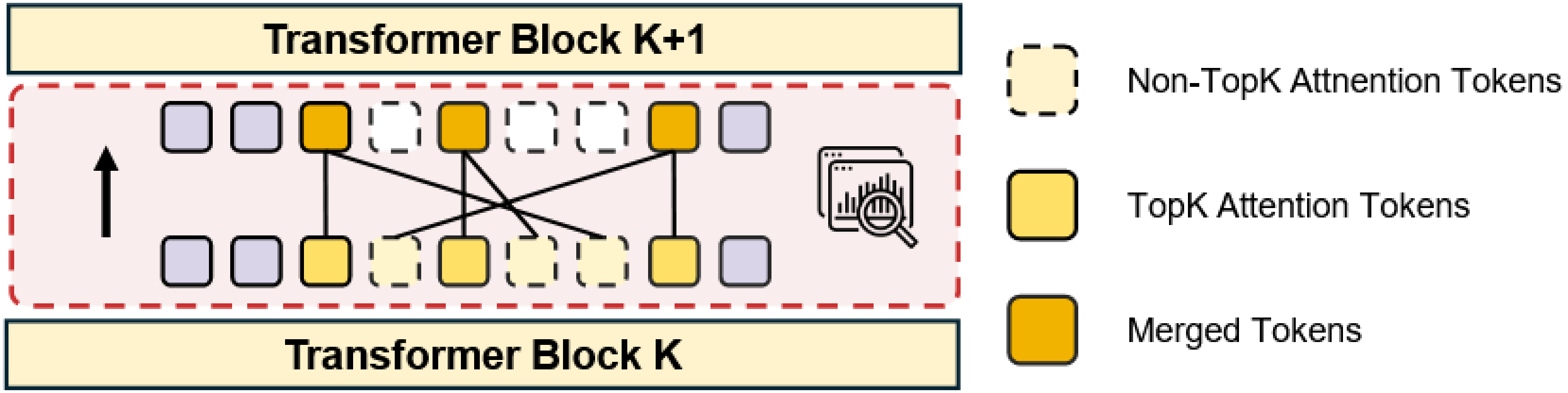
Non-Redundant:

- CLS token or compute the attention matrix $A \in \mathbb{R}^{B \times N_v \times N_v}$
- $A_{\text{avg}} \in \mathbb{R}^{B \times N_v}$ is higher means the token is more important

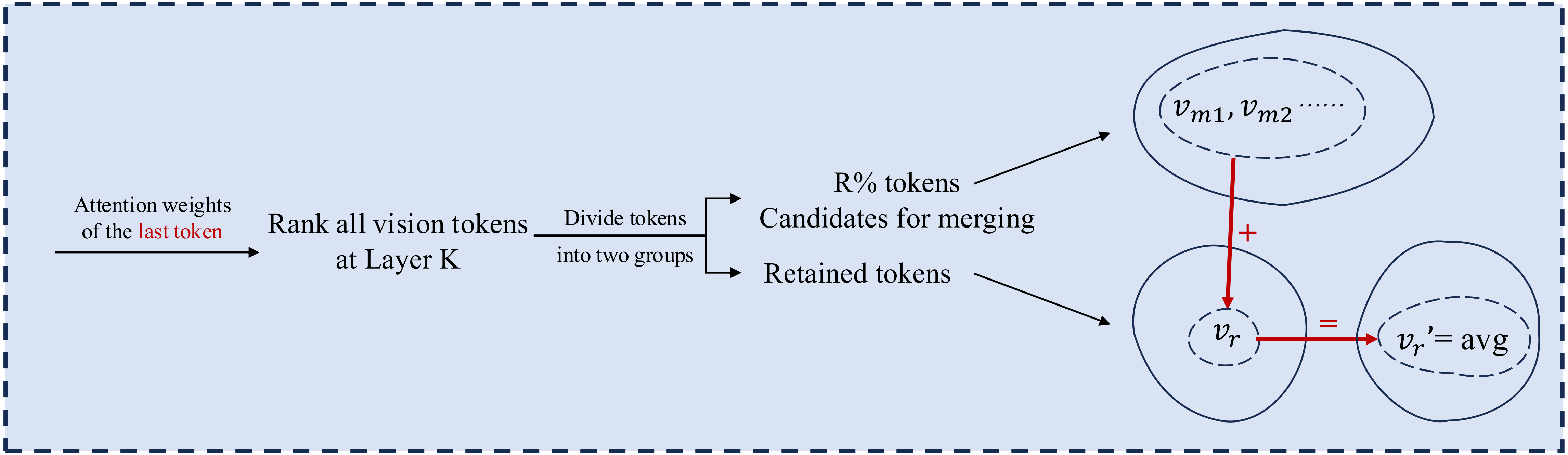
$$g(t_s, t_e) = \underbrace{\left(\sum_{k=1}^{N_v} \prod_{m=t_s}^{t_e-2} \mathbb{I}\big(sim(h_{m,k}, h_{m+1,k}) > \tau\big) \right)}_{N(t_s,t_e)} \times (t_e - t_s - 1)$$

$$\max_{K,\{t_i\}_{i=1}^{K+1}} \sum_{i=1}^{K} g(t_i, t_{i+1}) \longleftarrow \text{dp algorithm}$$

Inner-LLM



Transformer Block K+1
Transformer Block K
Non-TopK Attnention Tokens
TopK Attention Tokens
Merged Tokens



Attention weights
of the last token
Rank all vision tokens
at Layer K
Divide tokens
into two groups
R% tokens
Candidates for merging
Retained tokens
v_{m1}, v_{m2} ……
+
v_r
=
v_r'= avg

Results across benchmarks & backbones

Table 2: **Comparison of state-of-the-art methods across benchmarks. Best** and **most efficient** results are in bold, second best underlined. Here, "HoliTom" means the full version of our method; "HoliTom (w/o M)" means our method *without* inner-LLM merging, for reference.

Method	Prefilling FLOPs (T) ↓	FLOPs Ratio ↓	Before LLM Retained Ratio	MVBench ↑	EgoSchema ↑	LongVideo Bench ↑	VideoMME ↑	Avg. ↑ Score	%
LLaVA-OV-7B	40.8	100%	100%	58.3	60.4	56.4	58.6	58.4	100
FastV [6]	9.3	22.8%	100%	55.9	57.5	56.7	56.1	56.5	96.7
PDrop [53]	10.5	25.7%	100%	56.1	58.0	54.1	56.4	56.2	96.2
DyCoke [41]	8.7	21.3%	25%	53.1	59.5	49.5	54.3	54.1	92.6
VisionZip [57]	8.7	21.3%	25%	57.9	60.3	56.5	58.2	58.2	99.7
PruneVid [17]	8.7	21.3%	25%	57.4	59.9	55.7	57.4	57.6	98.6
FastVID [37]	8.7	21.3%	25%	56.5	-	56.3	58.0	-	-
HoliTom (w/o M)	8.7	21.3%	25%	**58.5**	60.8	56.5	**59.1**	58.7	100.5
HoliTom	**7.1**	**17.4%**	25%	58.4	**61.2**	**56.7**	58.9	**58.8**	**100.7**
VisionZip [57]	7.0	17.2%	20%	57.7	59.8	55.2	57.9	57.7	98.8
PruneVid [17]	7.0	17.2%	20%	57.2	59.7	54.7	56.9	57.1	97.8
FastVID [37]	7.0	17.2%	20%	56.3	-	**57.1**	57.9	-	-
HoliTom (w/o M)	7.0	17.2%	20%	58.5	60.7	56.3	**58.6**	58.5	100.2
HoliTom	**5.8**	**14.2%**	20%	**58.7**	**61.0**	**57.1**	**58.6**	**58.8**	**100.7**
VisionZip [57]	5.2	12.7%	15%	56.5	59.8	54.4	56.1	56.7	97.1
PruneVid [17]	5.2	12.7%	15%	56.8	59.7	55.4	56.6	57.1	97.8
FastVID [37]	5.2	12.7%	15%	56.0	-	56.2	57.7	-	-
HoliTom (w/o M)	5.2	12.7%	15%	**58.1**	61.0	**57.0**	**58.1**	**58.5**	**100.2**
HoliTom	**4.3**	**10.5%**	15%	**58.1**	**61.2**	56.4	57.3	58.2	99.7
VisionZip [57]	3.4	8.3%	10%	53.5	58.0	49.3	53.4	53.5	91.6
PruneVid [17]	3.4	8.3%	10%	56.2	59.8	54.5	56.0	56.6	96.9
FastVID [37]	3.4	8.3%	10%	55.9	-	56.3	**57.3**	-	-
HoliTom (w/o M)	3.4	8.3%	10%	**56.9**	61.1	**56.5**	56.9	57.8	99.0
HoliTom	**2.8**	**6.9%**	10%	**57.3**	**61.2**	56.3	56.8	**57.9**	**99.1**

Table 3: **Cross-backbone method comparison.** Performance comparison of our method against state-of-the-art methods across different backbones, demonstrating consistent effectiveness.

Model	Method	Prefilling FLOPs (T) ↓	FLOPs Ratio ↓	Before LLM Retained Ratio	MVBench ↑	EgoSchema ↑	LongVideo Bench ↑	VideoMME ↑	Avg. ↑ Score	%
LLaVA-OneVision-72B	Vanilla	429.3	100%	100%	60.9	61.1	62.7	65.7	62.6	100
	FastV [6]	86.4	20.1%	100%	56.1	57.1	57.0	61.2	57.9	92.5
	Visionzip [57]	59.0	13.7%	15%	58.4	59.3	57.4	63.8	59.7	95.4
	PruneVid [17]	59.0	13.7%	15%	56.8	57.7	57.4	62.7	58.6	93.6
	HoliTom (w/o M)	59.0	13.7%	15%	58.5	60.0	**57.8**	64.1	**60.1**	**96.0**
	HoliTom	**51.6**	**12.0%**	15%	**58.7**	**60.1**	57.2	**64.3**	**60.1**	**96.0**
LLaVA-Video-7B	Vanilla	80.2	100%	100%	60.4	57.2	58.9	64.3	60.2	100
	FastV [6]	17.1	21.3%	100%	54.3	54.1	55.0	58.8	55.6	92.4
	PDrop [53]	19.5	24.3%	100%	55.9	54.3	54.7	61.9	56.7	94.2
	VisionZip [57]	9.3	11.6%	15%	56.7	54.7	54.7	60.7	56.7	94.2
	HoliTom (w/o M)	9.3	11.6%	15%	**57.8**	**54.8**	55.6	61.9	57.5	95.5
	HoliTom	**7.6**	**9.5%**	15%	57.7	**54.8**	**56.2**	**62.1**	**57.7**	**95.8**

Ablation study & Efficiency study

Table 4: **Ablation study on merging modules.** Our temporal merging module reduces FLOPs to 75.7% without performance loss, alleviates the performance degradation caused by aggressive spatial pruning. The integration of all 3 modules achieves the best performance-efficiency trade-off.

Method	Prefilling FLOPs (T) ↓	FLOPs Ratio ↓	Before LLM Retained Ratio	MVBench ↑	EgoSchema ↑	LongVideo Bench ↑	VideoMME ↑	Avg. ↑ Score	%
Vanilla	40.8	100%	100%	58.3	60.4	56.4	58.6	58.4	100
Only Temporal	30.9	75.7%	79%	**58.9**	60.5	56.5	**59.1**	**58.8**	**100.7**
Only Spatial	5.2	12.7%	15%	57.9	60.8	54.2	56.8	57.4	98.3
HoliTom (w/o M)	5.2	12.7%	15%	58.1	61.0	**57.0**	58.1	58.5	100.2
HoliTom	**4.3**	**10.5%**	15%	58.1	**61.2**	56.4	57.3	58.2	99.7

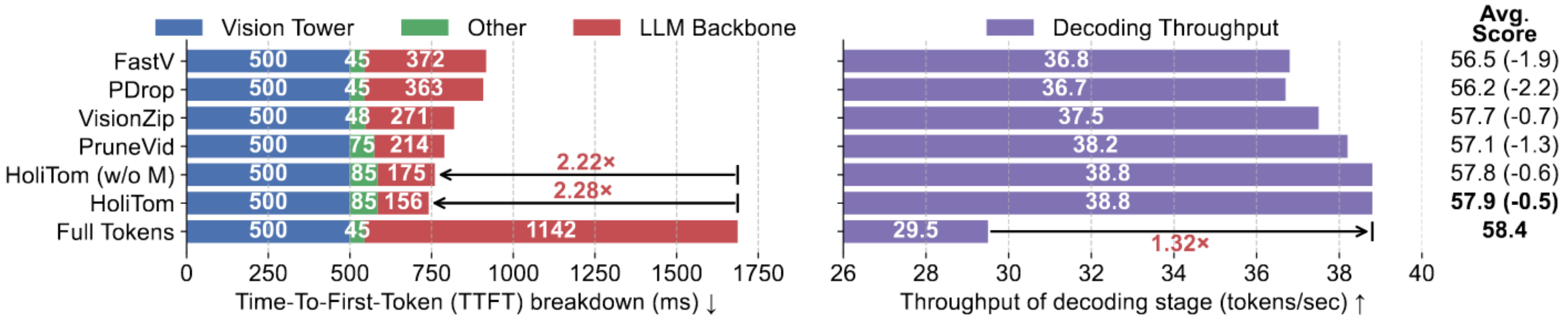
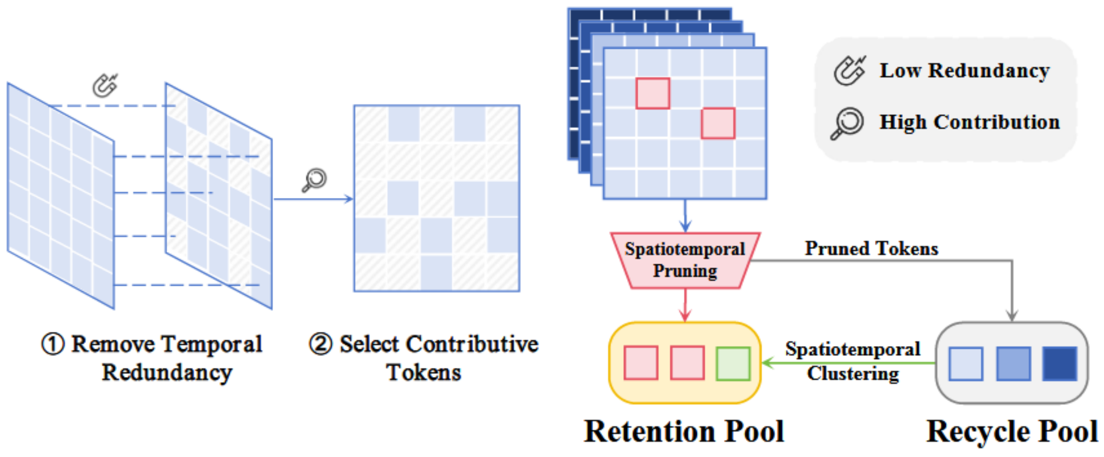


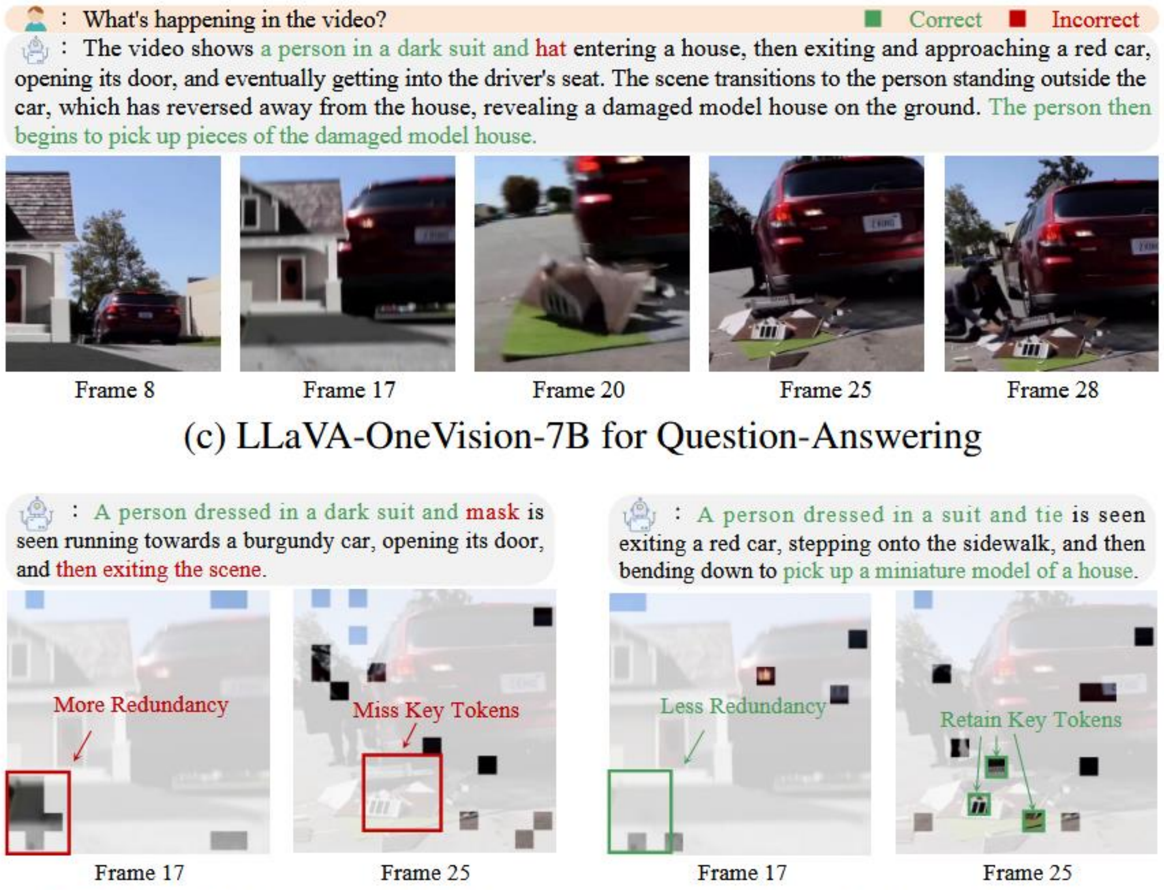
Figure 5: **Achieving superior inference.** "Other" indicates token pre-processing time (e.g., pooling). Our proposed method reduces Time-To-First-Token (TTFT) by 2.28× and achieves 1.32× higher decoding throughput, outperforming all other token compression methods and the vanilla model.

Unified Spatiotemporal Token Compression for Video-LLMs at Ultra-Low Retention



(a) Two-Stage Token Compression (b) Unified Spatiotemporal Compression

Comparison of methods' architecture



(c) LLaVA-OneVision-7B for Question-Answering

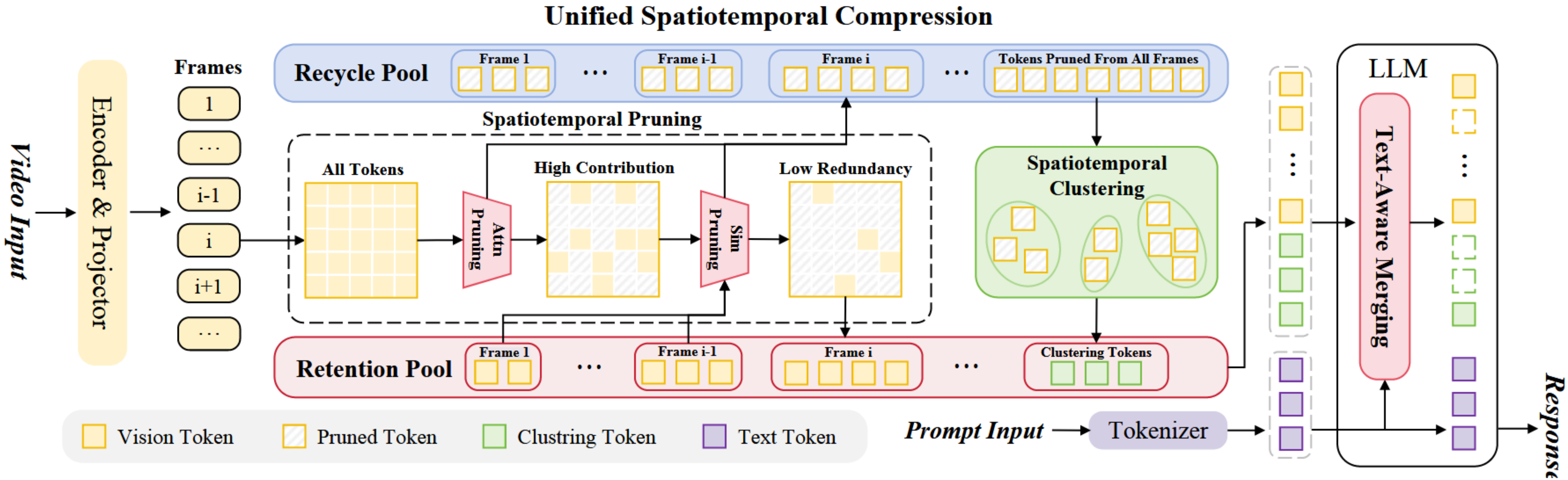
(d) Selected Tokens by HoliTom (e) Selected Tokens by Ours

Comparison of methods' performance

- Previous methods implicitly assuming that spatiotemporal redundancy in video tokens is separable
- Last token not only introduces positional bias but also diminishes the semantic influence of key query terms

Motivation

Architecture



Attn Pruning

CLS Token

Attention matrix

Cosine similarity

$$\boldsymbol{S} = \mathrm{sim}(c, \mathcal{P}) = \max_{p \in \mathcal{P}} \frac{c \cdot p}{\|c\|\|p\|}$$

$\boldsymbol{S} < \tau \longrightarrow$ Retention Pool

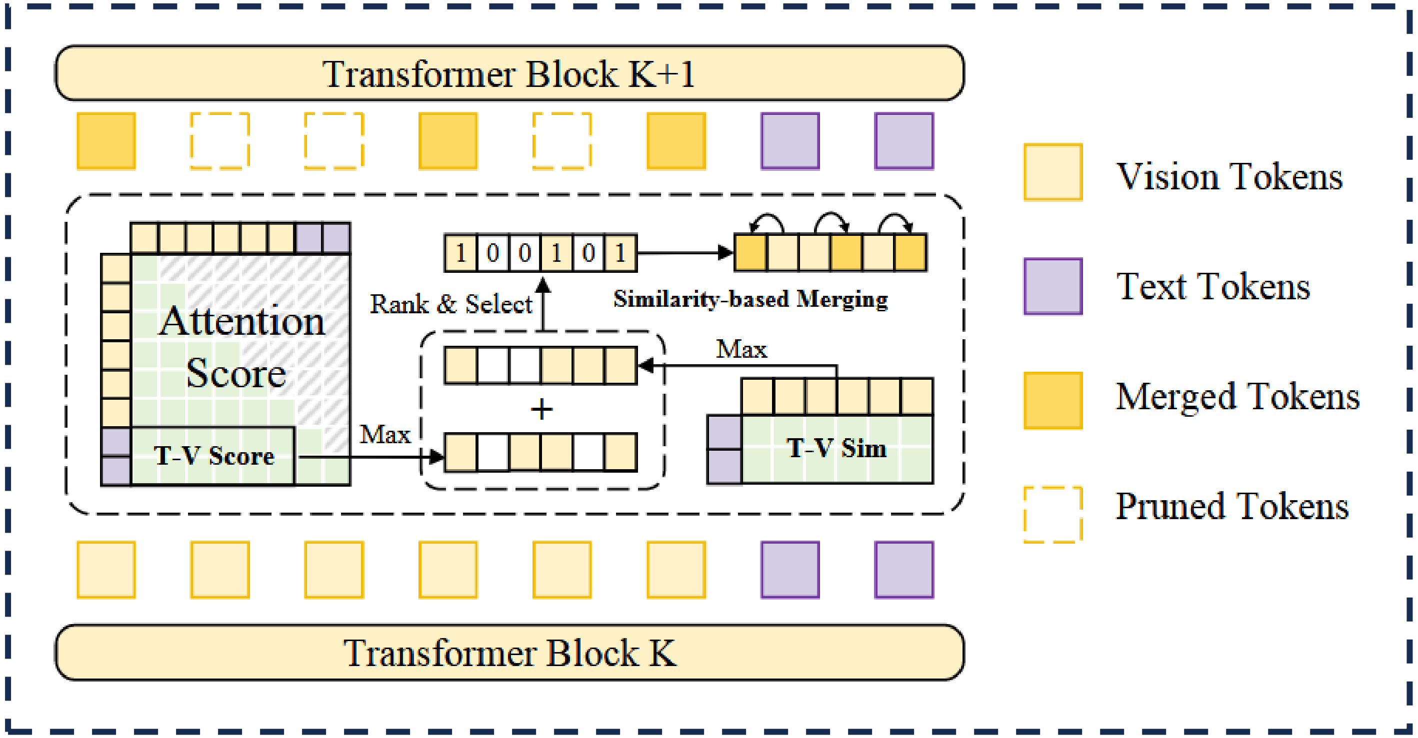
$\boldsymbol{S} > \tau \longrightarrow$ Recycle Pool

Spatiotemporal Clustering

DPC-KNN

Rearranged strictly according to their original spatiotemporal ordering

Text-aware Merging



- T-V Sim: compute semantic similarity between vision token and text token

$$S_m(v_i) = \max_{t \in \mathcal{T}} \frac{v_i \cdot t}{\|v_i\| \|t\|}$$

$$S_m^{\text{norm}} = \frac{S_m - \min(S_m)}{\max(S_m) - \min(S_m)}$$

- T-V Score: find which vision token is focused by text token

$$A_{qv} = A[N_v' :, : N_v']$$

$$A_m[j] = \max_{i=1}^{N_q} A_{qv}(i, j)$$

$$A_m^{\text{norm}} = \frac{A_m - \min(A_m)}{\max(A_m) - \min(A_m)}$$

- Rank、Select & Merge

$$I(v_i) = (1 - \lambda) \cdot A_m^{\text{norm}}(v_i) + \lambda \cdot S_m^{\text{norm}}(v_i)$$

$$r(j) = \arg \max_{v_k \in \mathcal{V}_{\text{retaining}}} \frac{v_j \cdot v_k}{\|v_j\| \|v_k\|}$$

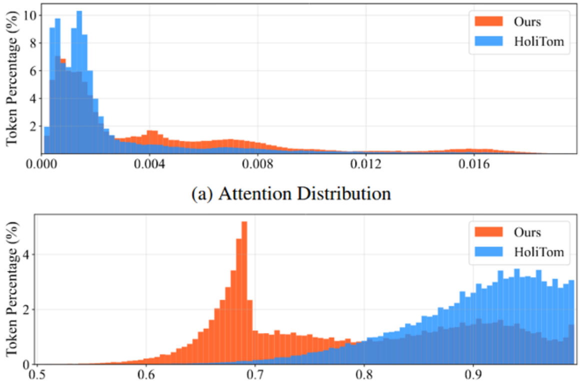
Results

Table 1. **Comparison of State-of-The-Art Methods on LLaVA-OneVision-7B.** The A%/B% retention ratio indicates that A% of the LLM input tokens are retained, and subsequently compressed to B% during the LLM forward pass. **Best** results are in bold, second best underlined. "(w/o M)" means our method without inner-LLM merging.

Method	FLOPs (T) ↓	FLOPs Ratio ↓	Retention Ratio	MVBench ↑	EgoSchema ↑	MLVU ↑	LongVideo Bench ↑	VideoMME ↑	Avg. ↑ Score	Avg. ↑ %
LLaVA-OV-7B [11]	41.4	100%	100%	58.3	60.4	47.7	56.4	58.6	56.3	100
FsatV [4]	7.9	19.1%	100%/10%	53.2	55.9	41.6	52.1	52.7	51.1	90.8
VisionZip [38]	4.0	9.6%	10%	53.5	58.0	42.5	49.3	53.4	51.3	91.2
LLaVA-Scissor [27]	4.0	9.6%	10%	-	57.5	-	-	55.8	-	-
FastVID [25]	4.0	9.6%	10%	55.9	58.7	42.6	56.3	**57.3**	54.2	96.2
HoliTom [23]	3.4	8.2%	10%/5%	57.3	**61.2**	45.1	56.3	56.8	55.3	98.3
Ours (w/o M)	4.0	9.6%	10%	57.4	60.5	44.7	**57.4**	56.6	55.3	98.3
Ours	3.4	8.2%	10%/5%	**57.7**	60.6	**45.5**	56.4	56.6	**55.4**	**98.4**
FastV [4]	6.4	15.5%	100%/5%	51.2	53.9	35.8	47.9	49.7	47.7	84.7
VisionZip [38]	2.2	5.4%	5%	45.2	51.9	37.4	46.4	48.2	45.8	81.4
LLaVA-Scissor [27]	2.2	5.4%	5%	-	56.6	-	-	53.3	-	-
FastVID [25]	2.2	5.4%	5%	53.0	57.1	42.1	51.1	54.2	51.5	91.5
HoliTom [23]	2.0	4.7%	5%/2.5%	55.6	**60.5**	40.6	53.6	54.2	52.9	94.0
Ours (w/o M)	2.2	5.4%	5%	**56.4**	59.5	42.5	53.7	**55.0**	53.4	94.9
Ours	2.0	4.7%	5%/2.5%	**56.4**	60.2	**42.8**	**54.5**	54.7	**53.7**	**95.4**
FastV [4]	5.5	13.3%	100%/2%	49.0	50.6	34.1	47.1	47.3	45.6	81.0
VisionZip [38]	1.2	2.9%	2%	41.7	47.6	31.8	45.1	45.9	42.4	75.3
FastVID [25]	1.2	2.9%	2%	48.0	52.3	37.6	47.3	49.2	46.9	83.3
HoliTom [23]	1.1	2.6%	2%/1%	52.6	57.2	37.4	48.5	51.1	49.4	87.7
Ours (w/o M)	1.2	2.9%	2%	**52.9**	57.2	39.5	**51.0**	51.3	50.4	89.5
Ours	1.1	2.6%	2%/1%	52.8	**57.6**	**40.3**	50.8	**51.8**	**50.7**	**90.1**
FastV [4]	5.2	12.6%	100%/1%	48.2	48.8	32.3	45.5	46.2	44.2	78.5
VisionZip [38]	0.9	2.1%	1%	40.8	43.8	29.7	44.4	44.3	40.6	72.1
FastVID [25]	0.9	2.1%	1%	45.3	47.8	32.4	46.1	47.0	43.7	77.7
HoliTom [23]	0.8	2.0%	1%/0.5%	49.6	52.9	33.9	48.1	49.0	46.7	82.9
Ours (w/o M)	0.9	2.1%	1%	**50.5**	53.3	**34.4**	**49.1**	**49.8**	**47.4**	**84.2**
Ours	0.8	2.0%	1%/0.5%	**50.5**	**53.8**	**34.4**	48.8	49.2	47.3	84.1

Table 2. **Cross-backbone Method Comparison.** Performance comparison of our method against state-of-the-art methods across different backbones, demonstrating consistent effectiveness.

Method	FLOPs (T) ↓	FLOPs Ratio ↓	Retention Ratio	MVBench ↑	EgoSchema ↑	MLVU ↑	LongVideo Bench ↑	VideoMME ↑	Avg. ↑ Score	Avg. ↑ %
LLaVA-Video-7B [46]	80.9	100%	100%	60.4	57.2	53.3	58.9	64.3	58.8	100
FsatV [4]	10.2	12.6%	100%/2%	42.9	41.0	34.3	45.6	48.1	42.4	72.1
VisionZip [38]	1.5	1.9%	2%	43.8	39.3	33.5	45.8	48.4	42.2	71.8
FastVid [25]	1.5	1.9%	2%	48.3	45.7	37.3	49.3	53.6	46.8	79.6
HoliTom [23]	1.4	1.7%	2%/1%	**50.2**	46.5	39.9	**50.7**	55.3	48.5	82.5
Ours (w/o M)	1.5	1.9%	2%	49.8	46.7	**40.8**	49.8	55.8	48.6	82.7
Ours	1.4	1.7%	2%/1%	50.1	**46.8**	**40.8**	50.2	**56.2**	**48.8**	**83.0**
FsatV [4]	9.7	11.9%	100%/1%	42.2	38.8	31.1	45.0	46.0	40.6	69.1
VisionZip [38]	1.2	1.5%	1%	42.3	36.8	30.9	45.6	47.0	40.5	68.9
FastVid [25]	1.2	1.5%	1%	43.6	38.8	30.1	46.2	49.3	41.6	70.7
HoliTom [23]	1.1	1.4%	1%/0.5%	46.4	41.1	38.9	48.8	51.7	45.4	77.2
Ours (w/o M)	1.2	1.5%	1%	47.9	**44.8**	40.0	48.9	54.6	47.2	80.3
Ours	1.1	1.4%	1%/0.5%	**48.0**	**44.8**	**40.1**	**49.4**	**54.9**	**47.4**	**80.6**
LLaVA-OV-0.5B [11]	3.54	100%	100%	46.6	26.6	33.5	47.5	43.7	39.6	100
FastV [4]	0.50	14.1%	100%/2%	39.2	21.2	22.5	38.8	34.0	31.1	78.5
VisionZip [38]	0.07	1.9%	2%	36.3	20.0	22.6	38.4	34.6	30.4	76.8
FastVid [25]	0.07	1.9%	2%	37.9	22.9	23.5	40.2	36.9	32.3	81.6
HoliTom [23]	0.06	1.8%	2%/1%	42.6	24.5	26.1	42.4	39.4	35.0	88.4
Ours (w/o M)	0.07	1.9%	2%	**43.0**	24.6	**26.4**	**45.3**	**40.5**	**35.9**	**90.7**
Ours	0.06	1.8%	2%/1%	42.7	**24.8**	**26.4**	45.2	40.4	**35.9**	**90.7**
FastV [4]	0.48	13.7%	100%/1%	35.6	19.3	21.5	37.9	32.2	29.3	74.0
VisionZip [38]	0.05	1.3%	1%	35.3	19.1	19.1	39.0	33.3	29.2	73.7
FastVid [25]	0.05	1.3%	1%	36.2	20.8	20.8	40.6	35.0	30.7	77.5
HoliTom [23]	0.05	1.3%	1%/0.5%	41.5	23.0	**25.1**	42.5	37.5	33.9	85.6
Ours (w/o M)	0.05	1.3%	1%	41.8	23.2	23.2	43.4	38.2	34.0	85.9
Ours	0.05	1.3%	1%/0.5%	**42.2**	**24.2**	23.1	**44.1**	**40.6**	**34.8**	**87.9**

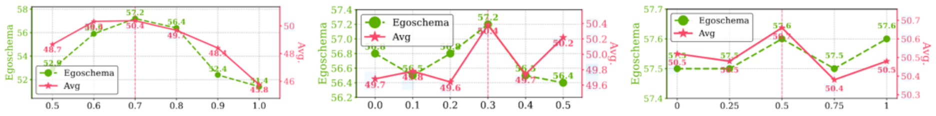


(a) Attention Distribution

(b) Semantic Similarity Distribution

Table 5. Module Ablations for Unified Spatiotemporal Token Compression.

Attn	Sim	Cluster	MVBench	EgoSchema	MLVU	LongVideo Bench	VideoMME	Avg. ↑ Score	Avg. ↑ %
✓			41.4	46.6	30.8	44.7	45.3	41.8	74.2
	✓		51.4	55.1	38.8	50.0	50.9	49.2	87.5
		✓	52.2	55.0	36.3	48.4	51.3	48.6	86.4
✓	✓		52.1	56.8	38.4	49.9	51.2	49.7	88.2
✓		✓	48.3	51.4	33.2	46.4	49.5	45.8	81.3
	✓	✓	51.8	55.5	38.4	50.6	51.0	49.5	87.9
✓	✓	✓	**52.9**	**57.2**	**39.5**	**51.0**	**51.3**	**50.4**	**89.5**



(a) Ablation on τ. (b) Ablation on Clustering Token Ratio. (c) Ablation on λ.

Figure 5. Ablation Experiment Results for Each Parameter.

Table 6. **Comparison of Efficiency and Performance.** Process, Prefill, and Time to First Token (TTFT) are measured in milliseconds; throughput is reported in tokens per second (token/s).

Method	Process	Prefill	TTFT	Throughput	Score
Vanilla [11]	-	701.2 (1×)	1104.7	27.6	56.3
VisionZip [38]	35.8	42.1 (17×)	451.6	35.4	45.8
FastVid [25]	**8.6**	43.1 (16×)	**446.2**	33.8	46.9
HoliTom [23]	88.7	40.5 (17×)	497.4	34.3	49.4
Ours	74.0	**31.3 (22×)**	473.8	**35.7**	**50.7**

When Token Pruning is Worse than Random:
Understanding Visual Token Information in VLLMs

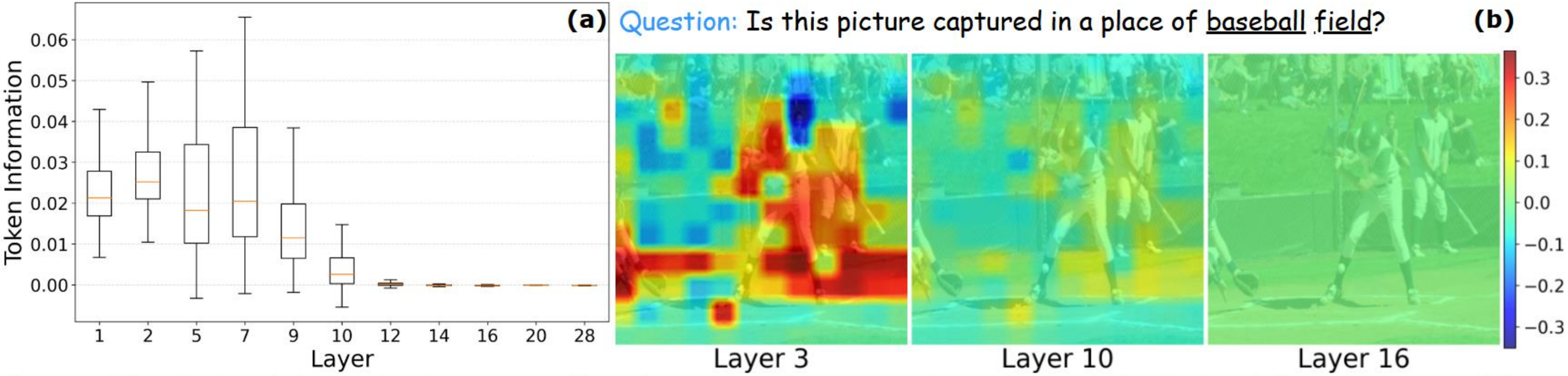


Figure 7. **Visual token information becomes uniform in the deep layers.** (a) The variance of visual token information across different layers of the language decoder. (b) Layer-wise visual token information maps at layers 3, 10, and 16.

Thanks for listening!

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